

Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

- A. Autoimmune hemolytic anemia
- B. Microangiopathic hemolytic anemia
- C. Paroxysmal nocturnal hemoglobinuria (PNH)
- D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

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The patient is a 35-year-old African American man with a history of recent anti-malaria prophylaxis prior to traveling to Nigeria. He presents with fever, abdominal pain, severe weakness, jaundice, generalized pallor, and laboratory findings of bite cells and Heinz bodies on peripheral smear.

Key clues:

- Bite cells and Heinz bodies are classic findings in G6PD deficiency, especially when triggered by drugs or infections (such as malaria).
- The patient received anti-malaria prophylaxis, which is a known trigger for hemolysis in G6PD deficiency.
- Normal G6PD enzyme levels in laboratory tests may seem contradictory, but in clinical practice, "normal" levels may not rule out functional deficiency or may refer to a different testing method (e.g., functional assay vs. genetic testing). However, the presence of bite cells and Heinz bodies strongly suggests hemolysis due to oxidative stress, which is characteristic of G6PD deficiency.
- Autoimmune hemolytic anemia (A) typically presents with spherocytes on peripheral smear, not bite cells or Heinz bodies.
- Microangiopathic hemolytic anemia (B) is associated with schistocytes, not bite cells or Heinz bodies.
- Paroxysmal nocturnal hemoglobinuria (C) presents with fragmented red cells and bone marrow failure, not typically bite cells or Heinz bodies.
- Although the lab shows normal G6PD levels, the clinical context (prophylaxis, bite cells, Heinz bodies) strongly suggests a functional deficiency or a drug-induced hemolysis due to G6PD deficiency.

Therefore, despite the normal lab test, the clinical picture and findings are most consistent with G6PD deficiency, which is a common cause of hemolytic anemia in African Americans and is triggered by drugs or infections like malaria.

<answer> Glucose - 6 -phosphate-dehydrogenase (G6PD) deficiency </answer>

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